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ABSTRACT

Natural disasters such as landslides, floods, and wildfires cause significant damage to infrastructure and human lives, necessitating rapid identification of affected regions for timely response. Traditional computer vision techniques often focus on prominent objects, overlooking smaller yet crucial elements indicative of disasters. To address this limitation, we propose the Retarget Network (RetNet), a novel deep learning framework designed to enhance image captioning models by shifting attention toward critical but less apparent disaster-affected regions. RetNet employs anchor boxes to define and optimize target areas, masking certain parts of an image to improve model focus. This approach enables better identification of objects such as displaced soil, rising water levels, or fire outbreaks, which may otherwise be misclassified. Our research validates RetNet's effectiveness across multiple disaster scenarios using various image perspectives, including aerial, side-view, and shipborne imagery.

Experimental results demonstrate that RetNet significantly enhances disaster region detection accuracy, achieving 91.60% accuracy in landslide identification from side-view images and 87.50% from shipborne views. The model effectively captures disaster-related details through improved image captioning, surpassing traditional classification techniques. Additionally, our ablation studies highlight the robustness of the proposed approach, showing that optimized masking through anchor box selection enhances caption generation. The proposed methodology has strong potential for real-world disaster response applications, aiding in early detection and assessment. Future work will focus on refining attention mechanisms for even greater precision, especially in complex disaster scenes with multiple overlapping elements.

INTRODUCTION

Extended Reality (XR), which includes Augmented Reality (AR) and Virtual Reality (VR), is revolutionizing human-computer interaction by creating immersive digital environments. However, traditional XR systems often rely on pre-defined responses and lack real-time adaptability, limiting their effectiveness in dynamic and complex scenarios. The emergence of Large Language Models (LLMs), such as OpenAI's GPT series, offers a transformative opportunity to enhance XR systems. By integrating natural language processing, contextual reasoning, and multimodal data interpretation, LLMs enable XR environments to become more intelligent, adaptive, and interactive. This synergy between XR and LLMs has the potential to bridge the gap between users and digital environments, making interactions more fluid, responsive, and human-like.

This seminar explores how LLMs can enhance XR systems across three key areas: Perception and Situational Awareness, Knowledge Modelling and Reasoning, and Visualization and Interaction. These capabilities can significantly improve applications such as healthcare, safety training, and architectural design by providing real-time contextual insights, intelligent decision-making, and interactive user experiences. Additionally, ethical considerations, such as privacy, transparency, and inclusivity, must be addressed to ensure responsible deployment of these technologies. By leveraging the power of LLMs in XR, we can create next-generation human-centric XR systems that redefine how users engage with digital environments.

LITERATURE SURVEY

Paper and Author details	Description	Advantages	Disadvantages
Authors: Liu, P., Wei, Y., Wang, Q., Chen, Y., & Xie, J. Title: "Research on post-earthquake" Year: 2020	A qualitative study" delves into the resilience processes of adult survivors half a decade after the 2008 Wenchuan earthquake in China. Through semi-structured interviews with 26 survivors from four reconstructed communities, the researchers identified four central themes in the resilience journey	The research provides insights into how Chinese cultural values influence resilience, offering valuable information for culturally sensitive post-disaster interventions.	☐ The qualitative nature and specific cultural context may limit the applicability of findings to other populations or disaster scenarios.
Authors: Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. Title: "Language models are unsupervised multitask learners. OpenAI Blog, 1(8), 9." Year: 2019	The paper "Language Models are Unsupervised Multitask Learners" by Radford et al. (2019) introduces GPT-2, a significant advancement in natural language processing (NLP). GPT-2 is a transformer-based language model trained on a vast corpus of internet text, enabling it to perform various tasks such as translation, summarization, and question answering without explicit task-specific training. This zero-shot learning capability demonstrates the model's ability to generalize across different NLP tasks, reducing the need for labeled datasets and task-specific architectures.	GPT-2 can produce coherent and contextually relevant text, making it valuable for applications like content creation, storytelling, and dialogue systems.	Due to its large size, GPT-2 requires substantial computational power for both training and inference, making deployment challenging for organizations with limited resources.

Authors: Li, Y., Wang, P., Feng, Q., Ji, X., Jin, D., & Gong, J. Title: "Landslide detection based on shipborne images and deep learning models: A case study in the Three Gorges Reservoir Area in China. Landslides, 20(3), 547–558." Year: 2023	The study titled "Landslide detection based on shipborne images and deep learning models: a case study in the Three Gorges Reservoir Area in China" addresses the challenges of detecting landslides in the riparian zones of the Three Gorges Reservoir Area (TGRA). Traditional remote sensing methods often fall short due to data collection difficulties and lack of facade information, leading to subjective, time-consuming, and less accurate results. This research leverages shipborne photogrammetry combined with deep learning techniques, specifically convolutional neural networks (CNNs) enhanced by transfer learning, to improve landslide detection accuracy.	The advantages of this approach include the ability to capture high-resolution, horizontal-view images of steep terrains, which are often challenging to assess using traditional aerial or satellite imagery. Shipborne photogrammetry offers flexibility, is less affected by airspace restrictions and weather conditions, and incurs lower costs compared to aerial methods.	The effectiveness of shipborne photogrammetry is constrained by river navigation conditions, and the cameras' limited field of view may not capture extensive areas in a single pass. Additionally, the reliance on deep learning models necessitates substantial labeled datasets for training, and the complexity of these models can pose challenges in terms of interpretability and computational requirements.
Authors: Ofli, F., Imran, M., Qazi, U., Roch, J., Pennington, C., Banks, V. J., & Bossu, R. (Title: "Landslide detection in real-time social media image streams. arXiv preprint arXiv:2110.04080." Year: 2021	The paper introduces an innovative approach to identifying landslides by leveraging social media imagery and artificial intelligence. The authors developed a computer vision model capable of real-time detection of landslides from social media image streams. To achieve this, they curated a substantial dataset of over 11,000 images, meticulously labeled by experts, and conducted extensive training experiments to optimize the model's performance.	The primary advantage of this approach is its ability to provide timely and scalable monitoring of landslide events without relying on active citizen participation or costly sensor deployments. By harnessing the vast amounts of data available on social media platforms, the system can enhance global landslide susceptibility maps and support emergency response efforts.	The reliance on social media data means that detection accuracy may be influenced by the quality and relevance of user-generated content, which can vary widely. Additionally, the system's effectiveness contingent upon the public's tendency to share such events online, potentially leading to underreporting in regions with low social media usage or restricted internet access.

PROBLEM STATEMENT

The accurate detection of disaster-affected regions in images is a critical challenge in emergency response and infrastructure assessment. Traditional computer vision techniques, such as object detection and image classification, primarily focus on identifying major objects while often overlooking smaller but significant disaster indicators. Landslides, floods, and wildfires frequently present complex visual patterns, making it difficult for conventional models to distinguish between normal and disaster-affected areas. Additionally, many machine learning models rely on manually engineered features, which are time-consuming to develop and may not generalize well across different disaster scenarios. As disaster patterns evolve, static feature sets become ineffective, leading to increased false positives and false negatives in detection. This inefficiency hinders response teams in assessing damage severity and taking timely action, emphasizing the need for an automated and adaptable detection framework..

Another major challenge in disaster detection is the lack of interpretability in deep learning models. Many state-of-the-art systems function as black-box models, providing high accuracy but offering little insight into their decision-making process. In real-world disaster management, human operators need to understand why a particular region has been flagged as disaster affected to make informed decisions. Without clear explanations, emergency responders may struggle to prioritize rescue efforts, allocate resources effectively, and mitigate potential risks. The inability to interpret model predictions also creates challenges in verifying detection accuracy, particularly in cases where environmental conditions, such as heavy fog, shadows, or structural debris, obscure disaster regions.

OBJECTIVES

The seminar aims to address the critical challenges in detecting disaster-affected regions by introducing the Retarget Network (RetNet), a novel framework designed to enhance image captioning models through attention-shifting mechanisms. The primary objectives of this seminar are:

1. **To explore the limitations of traditional computer vision techniques in disaster detection:** Discuss how conventional methods often overlook smaller yet significant disaster indicators, leading to inefficiencies in emergency response and infrastructure assessment.
2. **To introduce RetNet's attention-shifting mechanism:** Explain how RetNet employs anchor boxes to optimize region selection and masking, directing the model's focus toward less prominent but essential disaster-related features, thereby improving detection accuracy.
3. **To present validation results across multiple disaster scenarios:** Share empirical findings demonstrating RetNet's effectiveness, including achieving 91.60% accuracy in landslide detection from side-view images and 87.50% from shipborne views.
4. **To emphasize the importance of model interpretability in disaster management:** Highlight how RetNet's design enhances the transparency of deep learning models, allowing emergency responders to understand and trust the system's decision-making process.
5. **To encourage future research on automated attention-shifting techniques:** Advocate for the development of more precise and interpretable disaster detection tools, leveraging frameworks like RetNet to enhance emergency response and infrastructure assessment.

By focusing on these objectives, the seminar aims to shed light on innovative approaches to disaster detection and the potential of advanced image captioning models in improving disaster response strategies.

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